R13

Code No: 114CS

JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY HYDERABAD
B.Tech II Year II Semester Examinations, March - 2022
DESIGN AND ANALYSIS OF ALGORITHMS
(Computer Science and Engineering)

Time: 3 Hours **Max. Marks: 75**

Answer any five questions
All questions carry equal marks
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1.a) Write about the pseudocode conventions for representing algorithm.
b) Explain about Binary search and derive the time complexity. [8+7]

2.a) Illustrate on Asymptotic Notations.
b) Give the algorithm for Quick Sort. [7+8]

3.a) Explain about Depth first Graph Traversals with an example and derive the time complexity.
b) Explain AND/OR Graphs. [8+7]

4.a) Explain in detail about the binary tree traversal algorithm.
b) What is a disjoint set? Give the representations for disjoint sets and explain simple find and simple union operations. [8+7]

5.a) Explain job sequencing with deadlines using an example.
b) Discuss 0/1 knapsack problem using dynamic programming. [7+8]

6.a) Explain about the Optimal Binary Search Trees using dynamic programming.
b) Discuss about all pairs of shortest path problem. [8+7]

7.a) Give general method for Branch and Bound and compare with backtracking.
b) Give backtracking solution to sum of subsets problem. [8+7]

8.a) Write notes on NP-hard and NP-complete problems.
b) Explain cook's theorem. [8+7]

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